

Assignment 2: Prompt Engineering

1. Introduction

Prompt engineering is the practice of designing and refining inputs (prompts) to guide large language models (LLMs) and generative AI systems toward producing accurate, relevant, and creative outputs. It is a critical skill in applied AI, as the quality of the prompt directly influences the quality of the response.

2. Importance of Prompt Engineering

- Enhances **accuracy** of AI outputs.
- Reduces **ambiguity** in responses.
- Enables **control** over style, tone, and format.
- Improves **efficiency** in solving tasks like summarization, translation, coding, and reasoning.

3. Types of Prompt Engineering Techniques

1. Zero-Shot Prompting

Definition: Asking the model to perform a task without examples.

Use Case: Quick queries like *“Translate this sentence into French.”*

Strength: Fast and simple.

Limitation: May produce inconsistent results for complex tasks.

2. Few-Shot Prompting

Definition: Providing a few examples in the prompt to guide the model.

Use Case: *“Here are two examples of sentiment classification. Now classify this sentence.”*

Strength: Improves reliability by showing patterns.

Limitation: Requires careful selection of examples.

3. Chain-of-Thought Prompting

Definition: Encouraging the model to reason step by step before giving the final answer.

Use Case: Math problems, logical reasoning, or multi-step decision-making.

Strength: Produces more accurate reasoning.

Limitation: Can be verbose and slower.

4. Role/Persona Prompting

Definition: Assigning the model a role or identity to shape responses.

Use Case: *"You are a financial advisor. Explain investment risks."*

Strength: Controls tone and perspective.

Limitation: May require fine-tuning for consistency.

5. Instruction-Based Prompting

Definition: Giving explicit, structured instructions.

Use Case: *"Summarize this article in three bullet points."*

Strength: Clear and predictable outputs.

Limitation: Overly rigid prompts may reduce creativity.

6. Contextual Prompting

Definition: Embedding background information or constraints in the prompt.

Use Case: *"Based on the following dataset description, generate SQL queries."*

Strength: Makes outputs domain-specific.

Limitation: Longer prompts may confuse the model.

4. Best Practices

- Be **clear and specific** in instructions.
- Use **examples** when possible.

- Break complex tasks into **steps**.
- Experiment with **different prompt styles**.
- Continuously **refine prompts** based on results.

5. Conclusion

Prompt engineering is not just about asking questions—it is about **strategically designing inputs** to harness the full potential of generative AI. By mastering different techniques, practitioners can achieve outputs that are accurate, creative, and tailored to specific needs.